



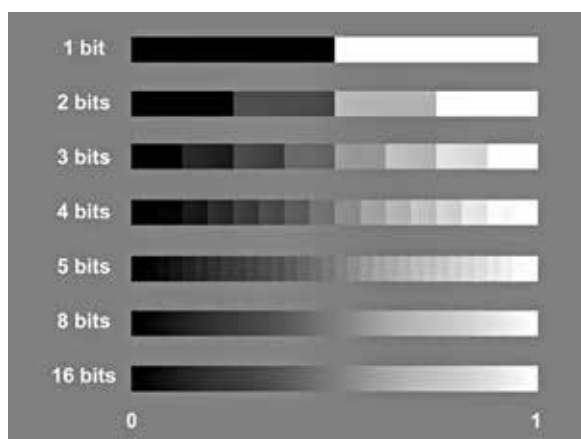
Creative Cloud on the Cloud!

Adobe is bringing stripped down versions of Photoshop and Illustrator to the web. <https://dgit.in/nov21-29>



Google halves its revenue share

Google will now take 15% share from developers as compared to 30% previously. <https://dgit.in/nov21-30>



0 and 1 and the transition from one extreme end to the other becomes more gradual and smooth. Look at the image above. You can see that 1 bit has an immediate transition from black to white. But 2-bit has 2 additional shades of grey in between black and white. Similarly, the number of shades

8-bits of Blue colour in it. This means that the Red channel can have 256 shades of red (2^8), the Green channel can have 256 shades of green and the Blue channel can have 256 shades of blue. Hence, the total number of colours it can make is $256 \times 256 \times 256 = 16$ Million Colours! Hence, 8-bit displays have become the standard for viewing our content. The average consumer monitor, tv and even smartphones have 8-bit displays. But of course, it doesn't stop there. People who work as professional video editors or photographers need displays which are of higher bit-depth. That's where 10-bit displays come in. Using the previous concepts, each channel can display 1024 shades of its colour. Hence, the total colours in a 10-bit panel is $1024 \times 1024 \times 1024 = 1.07$ Billion Colours!

CONCLUSION

There you have it! This article contains all the information that you need to know about colour spaces and colour gamuts in general. If you're reading this paragraph, then we're sure that you have a basic idea of how things work in the world of colour correction. Colour correction is an immensely complex field and we heartily salute those who do this for a living because this stuff gets way too complicated and technical

the points from the DCI P3 colour space to the sRGB colour space so that the final output looks similar to what it would've looked on a display that could display DCI P3 colour space? The answer is LUTs. LUT is an acronym for the word "Lookup Table". In a very simple definition, a LUT's job is to get the RGB value of one colour space as an input and generate an RGB value of another colour space as an output so that the input and the output look almost the same. Don't get us wrong here, LUTs are used for many different things too like colour grading, colour correction, etc. but since we're on the topic of colour spaces and colour gamuts, we decided to cite this example for a better understanding of the topic. You can see what a LUT does in essence in the picture on the previous page.

BIT-DEPTH

Whenever we go to buy a display of any sorts, we often see the word "8-bit panel" or "10-bit panel" thrown around pretty often. But what does "8-bit" or "10-bit" actually mean and how does it affect the colours on the screen? Well, we first need to understand what a "bit" means. A "bit" has two possible values. A 0 or a 1. So, if you consider 0 to be black and 1 to be white and you've got a 1 bit screen, that means that the screen can display either black or white.

Similarly, if you keep increasing the number of bits, you will end up with more shades of colour between

keep on increasing at the rate of 2^n where n is the number of bits.

The above paragraph was for black and white content. But what about colour content? It turns out that bit-depth for colour content isn't too different from the bit-depth of grayscale content. As we discussed earlier, an image is made up of RGB pixels. Now, each channel - Red, Green and Blue can be thought of as a band going from extreme left (0 - completely washed out - white) to the extreme right (1 - completely saturated colour) and we can implement the same concept as the above paragraph there too. Hence, if we take a 3 bit RGB (1 bit for Red, Green and Blue each), we end up an image that looks like this:



Source: <https://rgbtohex.page/models/color-bit-depth>



This above image originally looked like the image right next to it.

The important thing to understand here is that the bit-depth advertised everywhere is the BPC or Bits Per Channel. Hence, an 8-bit would have 8-bits of Red, 8-bits of Green and

way too fast. We hope that you learned something new from this article and if you have any doubts regarding any of the content mentioned in this article, feel free to shoot your doubts at editor@digit.in and we'll be more than happy to solve it for you!